

MÓDULO 1 - EXTRA **NETWORKING API**

LENDENDO DE UMA API REST EM JAVA

CLIENTE

VIA CEP

GET <https://viacep.com.br/ws/01001000/json/>

```
{
  "cep": "01001-000",
  "logradouro": "Praça da Sé",
  "complemento": "lado ímpar",
  "unidade": "",
  "bairro": "Sé",
  "localidade": "São Paulo",
  "uf": "SP",
  "estado": "São Paulo",
  "regiao": "Sudeste",
  "ibge": "3550308",
  "gia": "1004",
  "ddd": "11",
  "siafi": "7107"
}
```

java.net

URL e URLConnection

JAVA 1.0

CLIENTE

URL e URLConnection

```
String viaCepURL = "https://viacep.com.br/ws/01001000/json/";
```

```
URL url = new URL(viaCepURL);
```

```
URLConnection urlConnection = url.openConnection();
```

*Do pacote java.net, desde o
Java 1.0*

```
InputStream inputStream = urlConnection.getInputStream();
```

Abre a conexão aqui

```
Scanner scanner = new Scanner(inputStream);
```

```
while(scanner.hasNextLine()) {
```

```
    String line = scanner.nextLine();
```

```
    System.out.println(line);
```

```
}
```

```
scanner.close();
```

*Usando Scanner como se fosse
qualquer outro InputStream*

Fechando conexão corretamente

```
String viaCepURL = "https://viacep.com.br/ws/01001000/json/";
```

```
Scanner scanner = null;
```

```
try {
```

```
    scanner = new Scanner(new URL(viaCepURL).openStream());
```

```
    while(scanner.hasNextLine()) {
```

```
        String line = scanner.nextLine();
```

```
        System.out.println(line);
```

```
    }
```

```
} finally {
```

```
    if (scanner != null) {
```

```
        scanner.close();
```

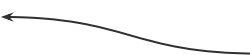
```
    }
```

```
}
```

*Abre o InputStream de
uma vez*



*Fechando Scanner tanto em caso
de sucesso como de exceção*



Try-with-resources

```
String viaCepURL = "https://viacep.com.br/ws/01001000/json/";
```

```
try (Scanner scanner = new Scanner(new URL(viaCepURL).openStream())) {  
    while (scanner.hasNextLine()) {  
        String line = scanner.nextLine();  
        System.out.println(line);  
    }  
}
```



*Fecha o InputStream de
maneira automática*

java.net.http

HttpClient

HttpClient

URI ao invés de URL

```
URI viaCepURI = URI.create("https://viacep.com.br/ws/01001000/json/");
```

```
try (HttpClient client = HttpClient.newBuilder().build()) {
```

Do pacote java.net.http

```
    HttpRequest request = HttpRequest.newBuilder().uri(viaCepURI).build();
```

```
    HttpResponse<String> response = client.send(request,  
                                                HttpResponse.BodyHandlers.ofString());
```

```
    System.out.println(response.statusCode());
```

```
    System.out.println(response.body());
```

*Conceitos HTTP: request,
response, status code, body*

```
}
```

com.sun.net.httpserver

jwebserver e HttpServer

Salvando itens do cardápio em JSON

```
Database database = new Database();  
List<ItemCardapio> itensCardapio = database.listaItensCardapio();  
  
Gson gson = new Gson();  
String json = gson.toJson(itensCardapio);  
Path arquivoItens = Path.of("itensCardapio.json");  
Files.writeString(arquivoItens, json);
```

Na Main

Arquivo

*itensCardapio.json
salvo na raiz do projeto*

```
[{"id":1,"nome":"Refresco do Chaves", "descricao":"Suco de limão  
que parece de tamarindo e tem gosto de groselha.",  
"categoria":"BEBIDAS", "preco":2.99}, {"id":2, "nome":"Sanduíche  
de Presunto do Chaves", "descricao":"Sanduíche de presunto  
simples, mas feito com muito amor.",  
"categoria":"PRATOS_PRINCIPAIS", "preco":3.50, "precoPromocional":  
2.99}, {"id":5, "nome":"Torta de Frango da Dona Florinda",  
"descricao":"Torta de frango com recheio cremoso e massa  
crocante.", ...}]
```

Subindo um jwebserver

\$ **jwebserver**

Binding to loopback by default. For all interfaces use "-b 0.0.0.0" or "-b ::".

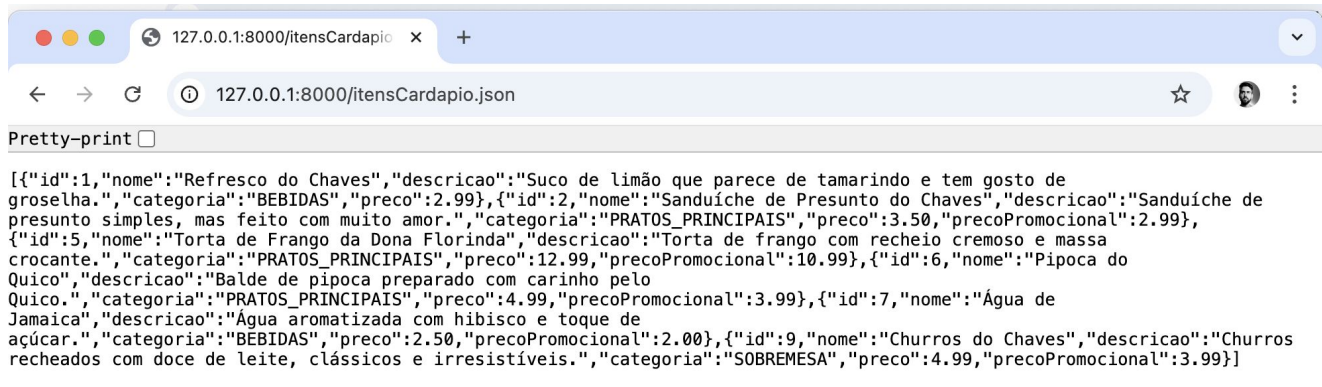
Serving /Users/alexandreaquiles/Documents/unipds/curso-modulo1-extra/cardapio and subdirectories on 127.0.0.1 port 8000

URL <http://127.0.0.1:8000/>



Directory listing for /

- [itensCardapio.json](#)
- [README.md](#)
- [gradle/](#)
- [gradlew](#)
- [build.gradle](#)
- [build/](#)
- [gradlew.bat](#)
- [settings.gradle](#)
- [src/](#)



Chamando o servidor com HttpClient

```
URI viaCepURI = URI.create("http://127.0.0.1:8000/itensCardapio.json");
```

Endereço do webserver

```
200
[{"id":1,"nome":"Refresco do Chaves","descricao":"Suco de limão que parece de tamarindo e tem gosto de groselha.","categoria":"BEBIDAS","preco":2.99}, {"id":2,"nome":"Sanduíche de Presunto do Chaves","descricao":"Sanduíche de presunto simples, mas feito com muito amor.","categoria":"PRATOS_PRINCIPAIS","preco":3.50,"precoPromocional":2.99}, {"id":5,"nome":"Torta de Frango da Dona Florinda","descricao":"Torta de frango com recheio cremoso e massa crocante.","categoria":"PRATOS_PRINCIPAIS","preco":12.99,"precoPromocional":10.99}, {"id":6,"nome":"Pipoca do Quico","descricao":"Balde de pipoca preparado com carinho pelo Quico.","categoria":"PRATOS_PRINCIPAIS","preco":4.99,"precoPromocional":3.99}, {"id":7,"nome":"Água de Jamaica","descricao":"Água aromatizada com hibisco e toque de açúcar.","categoria":"BEBIDAS","preco":2.50,"precoPromocional":2.00}, {"id":9,"nome":"Churros do Chaves","descricao":"Churros recheados com doce de leite, clássicos e irresistíveis.","categoria":"SOBREMESA","preco":4.99,"precoPromocional":3.99}]
```

Subindo um HttpServer

```
HttpServer httpServer = HttpServer.create(new InetSocketAddress(8000), 0);

httpServer.createContext("/itensCardapio.json", exchange -> {
    Path path = Path.of("itensCardapio.json");
    String json = Files.readString(path);

    byte[] bytes = json.getBytes();

    Headers responseHeaders = exchange.getResponseHeaders();
    responseHeaders.add("Content-Type", "application/json; charset=UTF-8");

    exchange.sendResponseHeaders(200, bytes.length);
    try(OutputStream responseBody = exchange.getResponseBody()){
        responseBody.write(bytes);
    }
});

System.out.println("Subiu servidor http! ");
httpServer.start();
```

Testando com cURL

```
$ curl -v 'http://localhost:8000/itensCardapio.json'
```

```
* Host localhost:8000 was resolved.
```

```
* IPv6: ::1
```

```
* IPv4: 127.0.0.1
```

```
* Trying [::1]:8000...
```

```
* Connected to localhost (::1) port 8000
```

```
> GET /itensCardapio.json HTTP/1.1
```

```
> Host: localhost:8000
```

```
> User-Agent: curl/8.7.1
```

```
> Accept: */*
```

```
>
```

```
* Request completely sent off
```

```
< HTTP/1.1 200 OK
```

```
< Date: Wed, 13 Aug 2025 00:17:02 GMT
```

```
< Content-type: application/json; charset=UTF-8
```

```
< Content-length: 1036
```

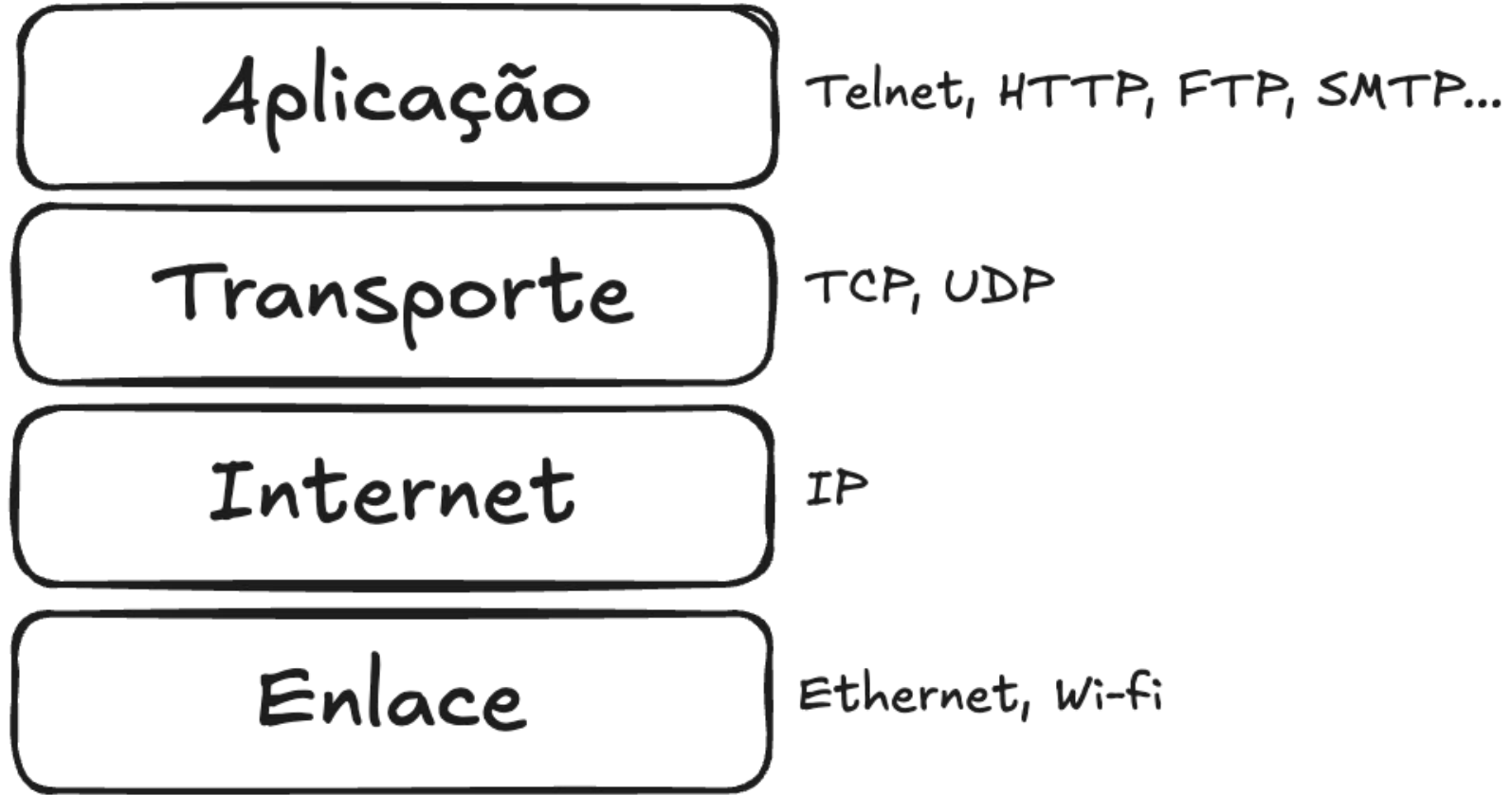
```
<
```

```
* Connection #0 to host localhost left intact
```

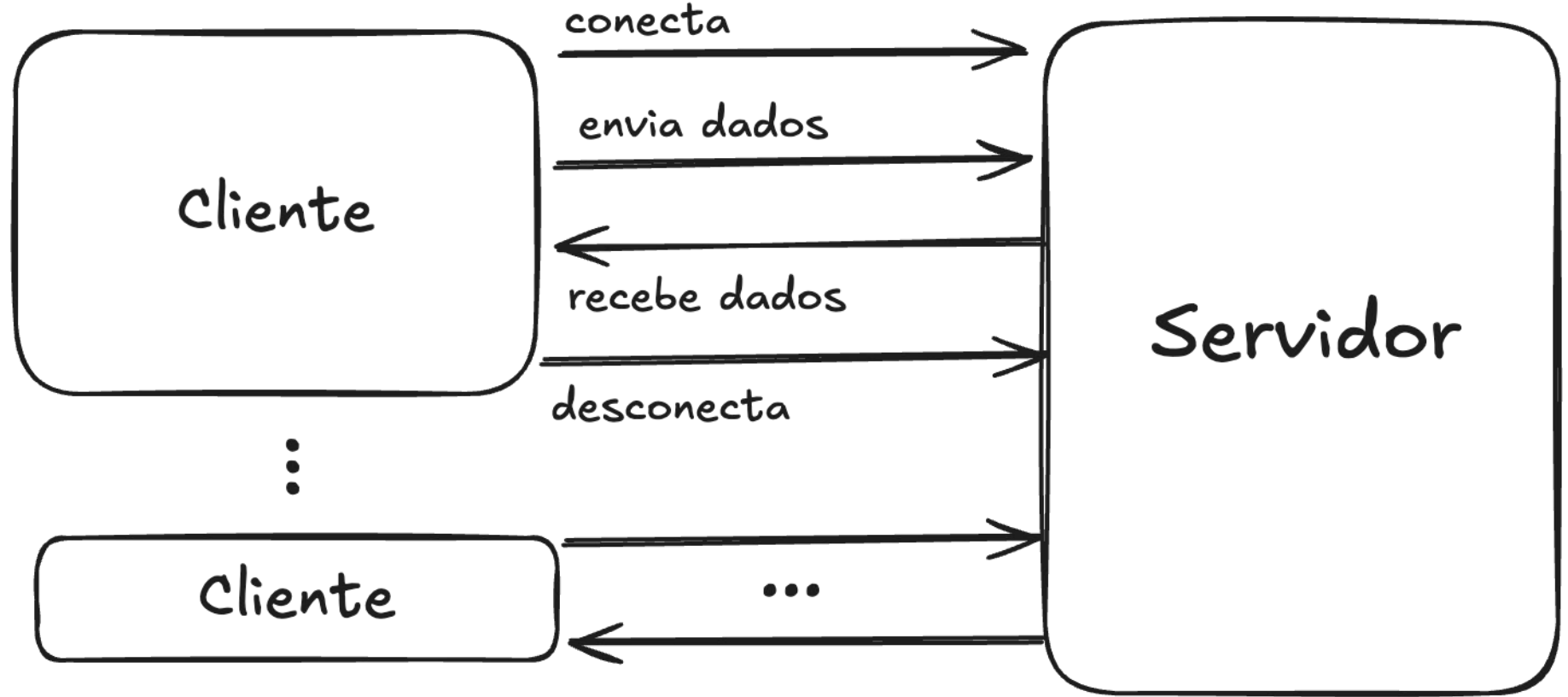
```
[{"id":1,"nome":"Refresco do Chaves","descricao":"Suco de limão que parece de tamarindo e tem  
gosto de groselha.", "categoria":"BEBIDAS", "preco":2.99}, {"id":2,"nome":"Sanduíche de Presunto  
do Chaves", "descricao":"Sanduíche de presunto simples...}]%
```

TCP/IP e Sockets

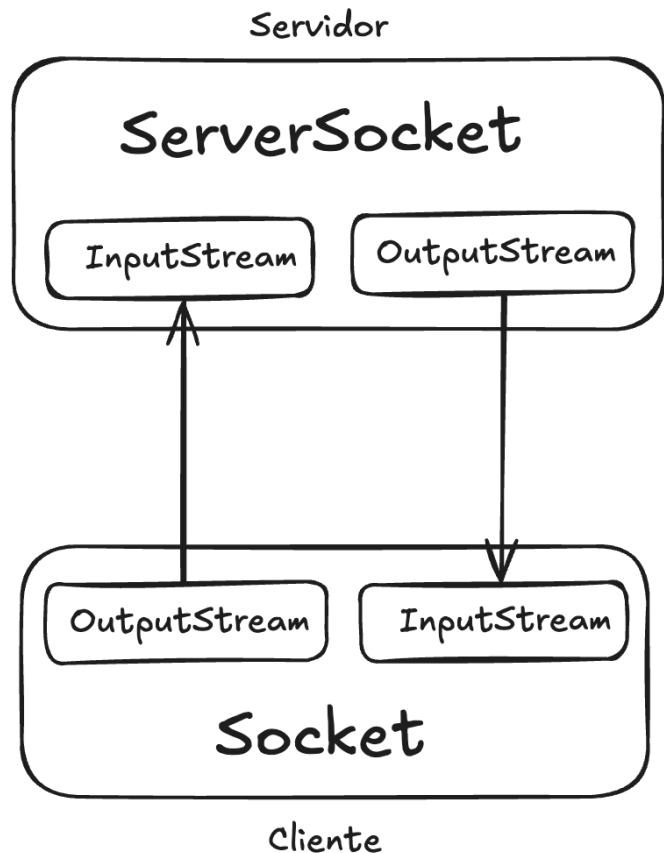
Modelo TCP/IP



Fluxo TCP



Sockets



endereço físico

IP

172.217.28.196 : 80

endereço da aplicação

Porta

www.google.com

RFC 2616: o protocollo HTTP/1.1

Request-Line = Method SP Request-URI SP HTTP-Version CRLF

Method = "OPTIONS" ; [Section 9.2](#)
| "GET" ; [Section 9.3](#)
| "HEAD" ; [Section 9.4](#)
| "POST" ; [Section 9.5](#)
| "PUT" ; [Section 9.6](#)
| "DELETE" ; [Section 9.7](#)
| "TRACE" ; [Section 9.8](#)
| "CONNECT" ; [Section 9.9](#)
| extension-method
extension-method = token

Request-URI = "*" | absoluteURI | abs_path | authority

GET /pub/WWW/TheProject.html HTTP/1.1

<https://datatracker.ietf.org/doc/html/rfc2616#section-5.1>

Testando com Telnet

```
$ telnet
```

```
telnet> open localhost 8000
```

Conectando ao servidor

```
Trying 127.0.0.1...
```

```
Connected to localhost.
```

```
Escape character is '^['.
```

```
GET /itensCardapio.json HTTP/1.1
```

*Solicitando o recurso
/itensCardapio.json*

```
HTTP/1.1 200 OK
```

```
Date: Sun, 10 Aug 2025 18:15:29 GMT
```

```
Content-type: application/json; charset=UTF-8
```

```
Content-length: 1036
```

```
[{"id":1,"nome":"Refresco do Chaves","descricao":"Suco de limão  
que parece de tamarindo e tem gosto de ...
```

Usando um ServerSocket

```
int porta = 8000;  
try(ServerSocket serverSocket = new ServerSocket(porta)) {  
    System.out.println("Iniciando servidor na porta " + porta);
```

*Cria
ServerSocket*

```
try (Socket clientSocket = serverSocket.accept()) {  
    System.out.println("Conectou com cliente " +  
        clientSocket.getInetAddress());
```

*Nova conexão com um
cliente*

```
    InputStream in = clientSocket.getInputStream();
```

```
    StringBuilder requestBuilder = new StringBuilder();
```

```
    int data;
```

```
    do {  
        data = in.read();  
        requestBuilder.append((char) data);  
    } while(in.available()>0);
```

```
    String request = requestBuilder.toString();
```

```
    System.out.println(request);
```

*Jeito mais baixo nível
de ler dados*

```
    }  
}
```

Testando o ServerSocket

← → ↻ ⓘ localhost:8000



This site can't be reached

localhost refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_REFUSED

Details

```

Iniciando servidor na porta 8000
Conectou com cliente /0:0:0:0:0:0:1
GET / HTTP/1.1
Host: localhost:8000
Connection: keep-alive
Cache-Control: max-age=0
sec-ch-ua: "Not;A=Brand";v="99", "Google Chrome";v="139",
"Chromium";v="139"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "macOS"
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0
Safari/537.36
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,i
mage/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=
0.7
Sec-Fetch-Site: none
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: en-US,en;q=0.9,pt;q=0.8

```

Response no ServerSocket

*Lista itens e
gera JSON*

```
Database database = new Database();  
List<ItemCardapio> listaItensCardapio = database.listaItensCardapio();  
Gson gson = new Gson();  
String json = gson.toJson(listaItensCardapio);
```

```
PrintStream out = new PrintStream(clientSocket.getOutputStream());  
out.println("HTTP/1.1 200 OK");  
out.println("Content-Type: application/json; charset=UTF-8");  
out.println("");  
out.print(json);
```

Envia o response

Testando com cURL

```
$ curl -v 'http://localhost:8000/itensCardapio.json'
```

```
* Host localhost:8000 was resolved.
```

```
* IPv6: ::1
```

```
* IPv4: 127.0.0.1
```

```
* Trying [::1]:8000...
```

```
* Connected to localhost (::1) port 8000
```

```
> GET /itensCardapio.json HTTP/1.1
```

```
> Host: localhost:8000
```

```
> User-Agent: curl/8.7.1
```

```
> Accept: */*
```

```
>
```

```
* Request completely sent off
```

```
< HTTP/1.1 200 OK
```

```
< Content-Type: application/json; charset=UTF-8
```

```
<
```

```
* no chunk, no close, no size. Assume close to signal end
```

```
* Closing connection
```

```
[{"id":1,"nome":"Refresco do Chaves","descricao":"Suco de limão que parece de tamarindo e tem  
gosto de groselha.","categoria":"BEBIDAS","preco":2.99},{ "id":2,"nome":"Sanduíche de Presunto  
do Chaves","descricao":"Sanduíche de presunto simples, mas feito com muito  
amor.","categoria":"PRATOS_PRINCIPAIS","preco":3.50,"precoPromocional":2.99},...]%
```

Socket do lado do Cliente

Cria Socket



```
try(Socket clientSocket = new Socket("localhost", 8000)) {  
    try (PrintStream out =  
        new PrintStream(clientSocket.getOutputStream())) {  
        out.println("GET /itensCardapio.json HTTP/1.1\n");  
    }  
}
```

Lendo do Socket no Cliente

```
try(Socket clientSocket = new Socket("localhost", 8000)) {  
    try (PrintStream out =  
        new PrintStream(clientSocket.getOutputStream());  
  
        Scanner in = new Scanner(clientSocket.getInputStream())) {  
  
        out.println("GET /itensCardapio.json HTTP/1.1\n");  
  
        while (in.hasNextLine()) {  
            String line = in.nextLine();  
            System.out.println(line);  
        }  
    }  
}
```

*Lendo do
InputStream*

*Imprime na saída padrão o
que for enviado do Servidor*